Cropsy: full system architecture

Farmer question > evidence-checked answer > visuals | English, Tamil, Tanglish

FARMER
Text | Voice | Photo | PDF
English, Tamil, Tanglish
Parcel id (e.g. TM-31)

FRONTEND (React + Vite)

- **Chat** — history, New chat, retry
- **Answer view** — text, steps, cautions, source
- **Media panel** — photo, diagram, chart, video
- **Map view** — 16 layers, satellite panel
- **Terrain shadow** — sunlight hours, cloud toggle
- **Upload** — photo / PDF, voice (mic)

ask →

← *media*

BACKEND (FastAPI, Python) /api/v1/query and /api/media

1 GATE — Blocks abuse, empty input

2 UNDERSTAND — Crop, topic, language, parcel (Nova Micro + rules)

3 PHOTO CHECK — Likely issue vs reference photos

4 TOOLS — Parcel, weather, soil, satellite, sunlight

5 ROUTER — Easy: Nova Micro
Compare: Nova Lite

6 RETRIEVE — Titan search: dense + keyword + crop boost

7 CALCULATOR — Dose x area: urea, DAP, MOP totals

8 WRITE — Ministral / Nova, evidence only

9 VERIFY (checks V1-V14) — Number in the source? Right unit and basis?
Right purpose? Dose tied to its own product?
Fail: retry once, then remove the figure
No evidence: helpful fallback, no invented number

retry

10 CONFIDENCE — Level, source page, conflict notes, region check

SAFETY CARD — Banned / restricted chemical: no dose, refer to KVK

TEXT ANSWER IS SHOWN FIRST > media is a separate call

11 MEDIA PLANNER (separate call: pictures never delay the answer)

Nova Micro picks from a closed menu (about 700 tokens); keyword rules are the fallback. Nothing fits: text only.

- **Diagram** — drawn by code: pH, drip, spacing
- **Photo** — 73 reviewed + pest, fertilizer
- **Chart** — weather, calculator, real data only
- **Map / NDVI** — satellite, field zones, sunlight
- **Video** — 224 verified TNAU / ICAR

KNOWLEDGE AND DATA

- **TNAU / ICAR guides** — text passages with page numbers
- **Titan embeddings** — 256-d vectors of those passages
- **Parcel data** — soil, crop, weather, satellite indices
- **16 satellite layers** — NDVI EVI SAVI NDRE NDWI NDMI ET SOC DEM LULC TCI ETCI hybrid colour-blind RVI RSM
- **Terrain + sun** — DEM 30 m > horizon > sun hours (NOAA)
- **Safety lists** — pesticide status, 3 recommendation conflicts
- **Photos + videos** — 73 reviewed + pest/fertilizer photos, 224 videos

safety check | media (2nd call) | frontend | safety card | dashed arrow: reads data / calls AWS

AWS (ap-south-1; SSO profile locally, role when deployed)

- **Bedrock: Nova Micro, Lite** — understand, route, media pick
- **Bedrock: Ministral 3B / 8B** — writes the answer (Tamil too)
- **Bedrock: Titan embeddings** — text search, image match
- **Textract** — reads PDFs and photos
- **Transcribe (streaming)** — voice to text
- **S3 bucket** — staging package, photos
- **Lambda + URL (planned)** — waiting on IAM permission